

		$= 24.0 \text{ J kg}^{-1}$
12	B	$\Delta U = m\Delta\phi \propto \Delta\phi$ $\frac{\Delta U_{BC}}{\Delta U_{AB}} = \frac{\Delta\phi_{BC}}{\Delta\phi_{AB}} \Rightarrow \frac{-5.0}{+20} = \frac{\phi_C + 3.0}{-3.0 + 7.0}$ $\Rightarrow \phi_C = -4.0 \text{ J kg}^{-1}$
13	A	When two objects A and B are placed in thermal contact, heat flows from the hotter object